

Mathematical Methods I

The Tangent Space

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1. Natural Coordinate Functions

In Euclidean space E^n , the **natural coordinate functions** $x_1, x_2, \dots, x_n$ assign to each point the value of one of its Cartesian coordinates. In E^3 , they are usually denoted by x , y , and z .

Definition 1.1 (Natural coordinate function)

Let $p = (p_1, p_2, \dots, p_n) \in E^n$. The i th natural coordinate function is the function

$$x_i : E^n \longrightarrow \mathbb{R}, \quad x_i(p) = p_i. \quad (1)$$

Therefore, every point can be written using the identity

$$p = (x_1(p), x_2(p), \dots, x_n(p)). \quad (2)$$

2. Balls, Spheres, and Open Sets

Differentiability is studied locally. Balls and open sets are used to make precise what it means to be “near a point.”

Definition 2.1 (Sphere and balls)

Given $a \in E^n$ and $r > 0$, the equation

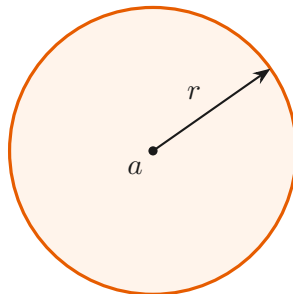
$$\|x - a\| = r \quad (3)$$

defines the **sphere** of radius r centered at a . The corresponding **open ball** and **closed ball** are, respectively,

$$B_r(a) = \{x \in E^n : \|x - a\| < r\}, \quad (4)$$

$$\overline{B}_r(a) = \{x \in E^n : \|x - a\| \leq r\}. \quad (5)$$

The sphere is the common boundary of both balls.



$B_r(a)$ and its boundary $\|x - a\| = r$

Figure 1: An open ball contains the points whose distance from the center is less than r .

Definition 2.2 (Open set)

A set $X \subseteq E^n$ is **open** if, for every $x \in X$, there exists a radius $r > 0$ such that

$$B_r(x) \subseteq X. \quad (6)$$

A point x belongs to the **boundary** of X if every open ball centered at x , no matter how small, contains at least one point of X and at least one point that does not belong to X . The set X is **closed** if it contains all its boundary points. A **neighborhood** of x is an open set containing x .

3. Differentiable Functions and Maps

Definition 3.1 (Differentiable function)

Let $X \subseteq E^n$ be an open set. A real-valued function $f : X \rightarrow \mathbb{R}$ is called **differentiable**, or of class C^∞ , if it has continuous partial derivatives of all orders.

In differential geometry, it is common to adopt the convention of working with C^∞ functions so that derivatives of any order can be computed without restriction. This is the notion of differentiability used in these notes.

Definition 3.2 (Differentiable map)

A map $F : X \subseteq E^n \rightarrow E^m$ can be written in terms of its coordinate functions as

$$F(p) = (f_1(p), f_2(p), \dots, f_m(p)). \quad (7)$$

The map F is **differentiable** if each function $f_j : X \rightarrow \mathbb{R}$ is of class C^∞ .

4. Tangent Vectors

A free vector describes only direction and magnitude. For differential calculus, it is also necessary to record the point in space at which it acts.

Definition 4.1 (Tangent vector)

A **tangent vector** to E^n is a pair consisting of a point of application $p \in E^n$ and a vector part $v \in E^n$. It is denoted by

$$v_p = (p, v). \quad (8)$$

Geometrically, it is represented by an arrow with its tail at p and with the direction and magnitude determined by v ; its tip is located at $p + v$.

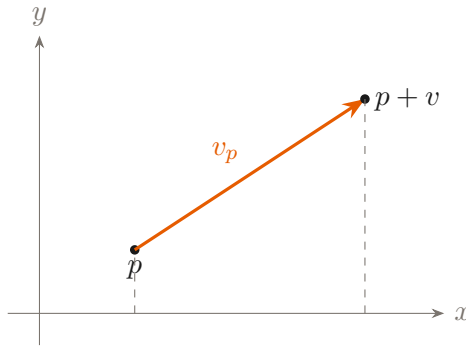


Figure 2: The tangent vector v_p is anchored at the point p .

Dependence on the point of application

Two tangent vectors with the same vector part but anchored at different points are distinct objects:

$$p \neq q \implies v_p \neq v_q. \quad (9)$$

Therefore, a tangent vector cannot be freely translated without changing it. This dependence on the point is essential when studying surfaces and manifolds.

5. The Tangent Space

Definition 5.1 (Tangent space)

For a fixed $p \in E^n$, the **tangent space** at p is the set of all tangent vectors anchored at that point:

$$T_p(E^n) = \{v_p : v \in E^n\}. \quad (10)$$

Because all its elements have the same point of application, the operations

$$v_p + w_p = (v + w)_p, \quad c v_p = (cv)_p, \quad (11)$$

can be defined, where $v, w \in E^n$ and $c \in \mathbb{R}$. With these operations, $T_p(E^n)$ is a vector space.

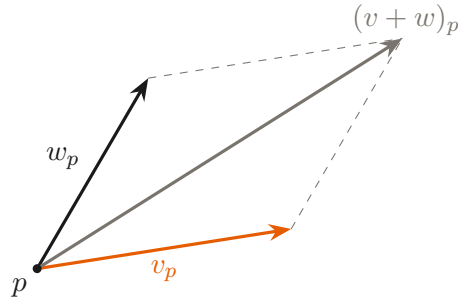


Figure 3: Addition of vectors within the same tangent space $T_p(E^n)$.

Proposition 5.2 (Canonical isomorphism)

For each fixed point p , the map

$$\Phi_p : E^n \longrightarrow T_p(E^n), \quad \Phi_p(v) = v_p, \quad (12)$$

is linear, injective, and surjective. Consequently,

$$T_p(E^n) \cong E^n, \quad \dim T_p(E^n) = n. \quad (13)$$

The isomorphism allows their algebraic structures to be identified, but it does not eliminate the point of application: v_p still belongs specifically to $T_p(E^n)$.

6. Vector Fields**Definition 6.1 (Vector field)**

A **vector field** V on an open set $X \subseteq E^n$ is an assignment that associates with each point $p \in X$ a tangent vector at that same point:

$$V : p \longmapsto V(p) \in T_p(E^n). \quad (14)$$

Thus, the point of application of $V(p)$ changes along with p .

7. The Natural Frame

In E^3 , consider three constant vector fields pointing in the positive directions of the Cartesian axes.

Definition 7.1 (Natural vector fields)

For each $p \in E^3$, define

$$U_1(p) = (1, 0, 0)_p, \quad U_2(p) = (0, 1, 0)_p, \quad U_3(p) = (0, 0, 1)_p. \quad (15)$$

The set (U_1, U_2, U_3) is called the **natural frame** or **natural reference frame** of E^3 .

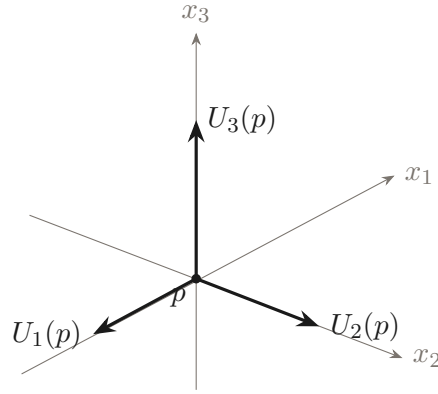


Figure 4: The natural frame assigns a basis to each tangent space.

At each point p , the vectors $U_1(p), U_2(p), U_3(p)$ form an orthonormal basis of $T_p(E^3)$. In E^n , the natural frame similarly consists of n fields $U_1, \dots, U_n$.

Proposition 7.2 (Canonical representation)

Every vector field V on E^n can be expressed uniquely as

$$V = \sum_{i=1}^n v_i U_i, \quad (16)$$

where $v_1, \dots, v_n : E^n \rightarrow \mathbb{R}$ are the **Euclidean coordinate functions** of V . Evaluating at a point gives

$$V(p) = \sum_{i=1}^n v_i(p) U_i(p). \quad (17)$$

For fields $V = \sum_i v_i U_i$ and $W = \sum_i w_i U_i$, and a real-valued function f , the operations are computed componentwise:

$$V + W = \sum_{i=1}^n (v_i + w_i) U_i, \quad (18)$$

$$fV = \sum_{i=1}^n (f v_i) U_i. \quad (19)$$

8. Differentiable Vector Fields

Definition 8.1 (Differentiable field)

Let

$$V = \sum_{i=1}^n v_i U_i \tag{20}$$

be a vector field on an open set $X \subseteq E^n$. The field V is **differentiable** if and only if each component function $v_i : X \rightarrow \mathbb{R}$ is of class C^∞ .